

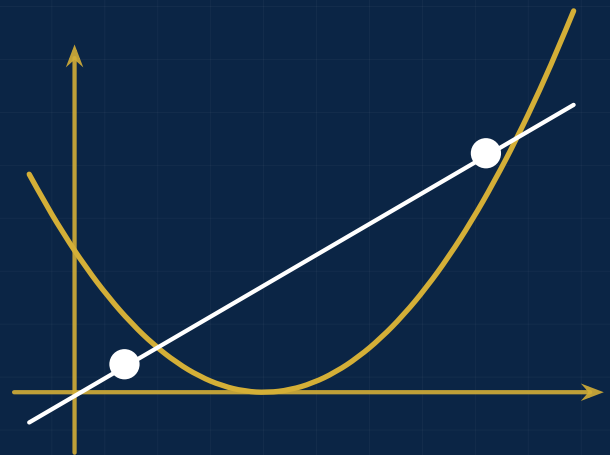


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CHAPTER 3 • SEQUENCES, FUNCTIONS & GRAPHS • TOPIC 5

Graphs of Functions

Table, plot, smooth curve. Then read the curve: solve equations with a line, name a shape from its equation, sketch a quadratic, decode a trig graph.



5

VARIANTS

40

PAST-PAPER PARTS

36

PRACTICE QUESTIONS

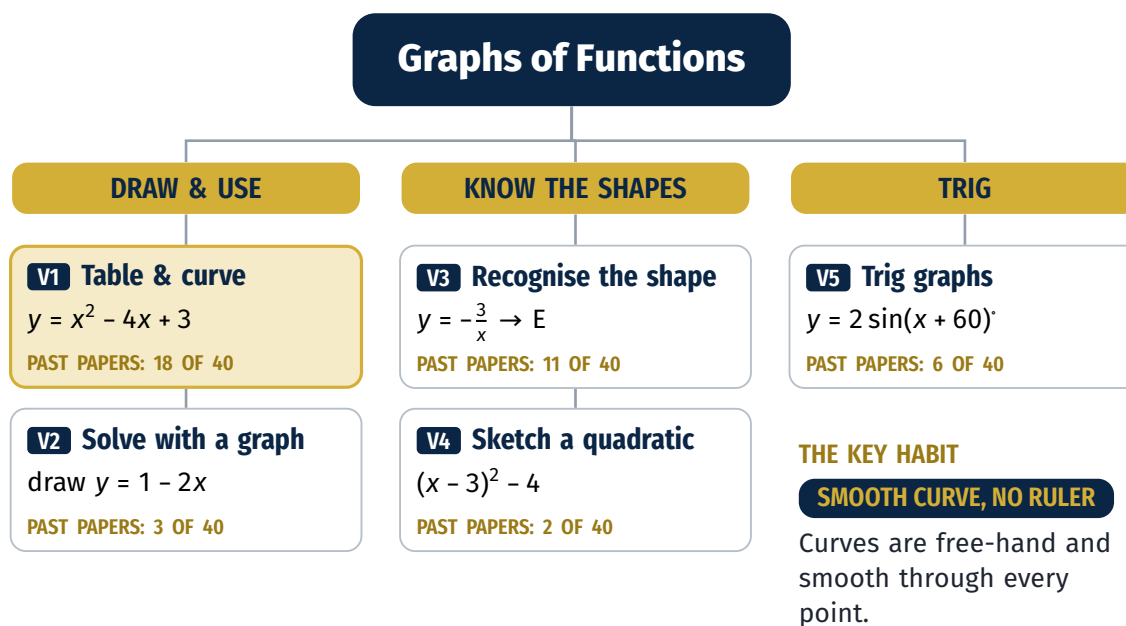
YOU WILL MASTER

- V1 Table & curve
- V2 Solve with a graph
- V3 Recognise the shape
- V4 Sketch a quadratic
- V5 Trig graphs

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Graphs of functions at a glance

 Your notes



★ How often does each variant come up?

22 past-paper questions (Jan 2020 → Nov 2025) hold 40 parts; 36 practice parts — grouped by type.

	PAST PAPERS (OUT OF 40)	PRACTICE PARTS
V1 Table, curve	18	13
V2 Solve with a graph	3	7
V3 Recognise the shape	11	3
V4 Sketch a quadratic	2	9
V5 Trig graphs	6	4

WHAT THIS TELLS YOU A “complete the table, draw the graph” question is on almost every paper (4 marks, early). A “which graph is which” question (3 marks) is the next most common. Both are pure technique.

BEFORE YOU START

The big idea

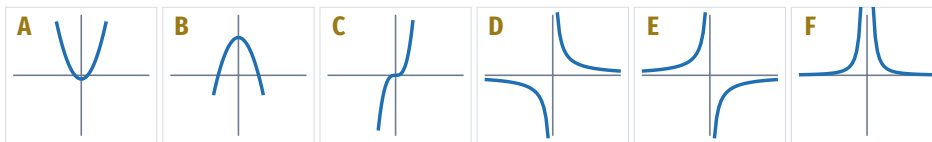
Every point on a graph solves its equation.

$(2, -1)$ is on $y = x^2 - 4x + 3$ because $4 - 8 + 3 = -1$.

So: a **table** gives points to draw; the graph gives **solutions** back; and each equation type has its own **shape**.



Know the shapes



A $y = x^2$ (U) B $y = -x^2$ (∩) C $y = x^3$ D $y = \frac{1}{x}$ E $y = -\frac{1}{x}$ F $y = \frac{1}{x^2}$



CALCULATOR FIRST, THEN CHECK

Table mode: $f(x) = x^2 - 4x + 3$, start -2, end 4, step 1 — the whole table at once. Check one value by hand.

 Your notes

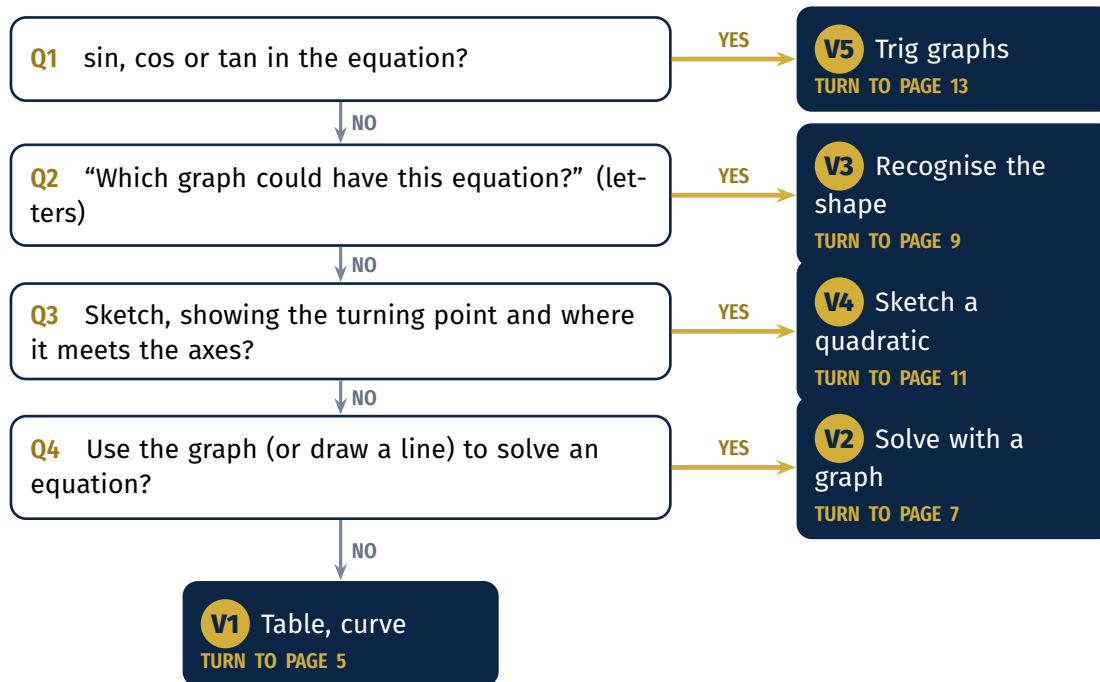
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1 MOST COMMON IN THE EXAM

Table & Curve

Work out each y carefully, plot, and join with one smooth curve.

SEEN IN PAST PAPERS

18 / 40

PRACTICE QUESTIONS: 13

 Your notes

Came up in: Jan 2020 P2HR Q15 • May 2021 P2H Q15 • Jan 2022 P1HR Q4 • Jan 2022 P2HR Q11 • Jun 2023 P2H Q4 • Nov 2023 P1H Q12 • Nov 2024 P1H Q11 • Jun 2025 P1H Q10 • Nov 2025 P2H Q12

? What is it?

In plain words

Fill the table: substitute each x —

bracket negatives: $(-2)^2 = 4$.

Plot each point with a small cross and join with **one smooth curve** (no ruler, no straight bits). A U-curve has a **rounded bottom**, not a point.

IT LOOKS LIKE

$$y = x^2 - 4x + 3$$

$$y = \frac{6}{x}$$

$$y = x^3 - 3x + 1$$

$$x = -2: (-2)^2 - 4(-2) + 3$$

$$= 4 + 8 + 3 = 15$$

$$\text{not } -4 + 8 + 3 \times$$

BRACKET THE NEGATIVE

How do I do it?

1 Table

Substitute each x ; bracket negatives (or use table mode).

2 Pattern

Quadratic tables are symmetric — use it to spot slips.

3 Plot

Small crosses, exactly.

4 Curve

One smooth curve through all points, over the full range.

SMOOTH CURVE, NO RULER

A point off the curve: recheck the table.

V1 Table & Curve

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Complete the table of values for $y = x^2 - 4x + 3$ ”

→ **Substitute; bracket negatives**

“On the grid, draw the graph ... for values of x from -2 to 4 ”

→ **Plot; smooth curve, whole range**

$y = \frac{6}{x}$, $y = \frac{1}{x}(x^2 + 4)$

→ **Reciprocal: never touches the axes****WORKED EXAMPLE****PAST PAPER**

January 2022 · Paper 1HR · Q4 · 4 marks

(a) Complete the table of values for $y = x^2 - 4x + 3$ for $x = -2, -1, 0, 1, 2, 3, 4$. (b) Draw the graph for x from -2 to 4 .

STEP 1 $x = -2 : 4+8+3 = 15$; $x = 1 : 1-4+3 = 0$ ◀ substitute

STEP 2 $y : 15, 8, 3, 0, -1, 0, 3$ ◀ (a) the table

STEP 3 symmetric about $x = 2$ ◀ check the pattern

STEP 4 Plot the 7 points; one smooth U-curve, lowest at $(2, -1)$ ◀ (b)

ANSWER 15, 8, 3, 0, -1, 0, 3; U-curve, minimum $(2, -1)$ ✓
symmetric ✓

**WATCH OUT**

✗ a pointed V at the bottom

✓ a rounded U

The curve turns smoothly between $x = 1$ and $x = 3$.

**EXAM TIP**

Turn the paper and draw from the inside of your wrist — smoother curves.

**Check yourself**

calculator first · answers on page 15

1.1

Jun 2023 P2H

$y = x^2 - x - 4$. Find y for $x = -3, -1, 0, 2, 3$.

Answer:

1.2

Jan 2022 P2HR

$y = \frac{6}{x}$. Find y for $x = 0.5, 2, 4, 5$.

Answer:

1.3

Nov 2023 P1H

$y = x^3 - 3x + 1$. Find y for $x = -2, -1, 0, 2$.

Answer:

V2

VARIANT 2

Solve Equations with a Graph

Rearrange the new equation into “curve = line”; draw the line; read the x values where they cross.

SEEN IN PAST PAPERS

3 / 40

PRACTICE QUESTIONS: 7



Your notes

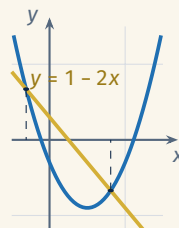
Came up in: Jan 2022 P2H Q21 • Nov 2025 P2H Q12(c)

? What is it?

In plain words

Solutions of “curve = 0” are where the curve crosses the **x-axis**.

For a different equation, make the curve’s expression appear on one side; the other side is a **straight line** to draw. The answers are the **x-values** of the crossings.



IT LOOKS LIKE

$$2x^2 - 4x - 1 = 0$$

$$x + \frac{4}{x} = 6$$

How do I do it?

1 Match

Rearrange so one side is the curve you have.

2 Line

The other side is the line to draw.

3 Draw

Rule the line on the grid.

4 Read

x-values of the crossings, to 1 d.p.

CURVE = LINE

Answers are x-values only.

V2 Solve Equations with a Graph

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Use the graph to find estimates for the solutions of $2x^2 - 4x - 1 = 0$ ”

→ **Where the curve crosses $y = 0$**

“By drawing a suitable straight line ... solve $x^2 - x - 1 = 0$ ”

→ **Make $2x^2 - 4x - 1$ appear**

“correct to one decimal place”

→ **Read carefully on the small squares****WORKED EXAMPLE** **PAST PAPER**

January 2022 • Paper 2H • Q21(b) • 3 marks

Part of the graph of $y = 2x^2 - 4x - 1$ is shown. By drawing a suitable straight line, find estimates for the solutions of $x^2 - x - 1 = 0$.

STEP 1 $2x^2 - 2x - 2 = 0$ ◀ $\times 2$ to get $2x^2$ **STEP 2** $2x^2 - 4x - 1 = -2x + 1$

◀ curve on the left

STEP 3 draw $y = 1 - 2x$

◀ the line

STEP 4 $x \approx -0.6, \quad x \approx 1.6$

◀ read the crossings

ANSWER $x \approx -0.6$ and $x \approx 1.6$ ✓ $1.6^2 - 1.6 - 1 = -0.04 \approx 0$ ✓**WATCH OUT**✗ giving the y -values✓ give the x -values

The solutions are the x -coordinates of the crossings.

**EXAM TIP**

Write the rearrangement line — it is the method mark, even if your reading is a little off.

**Check yourself**

calculator first • answers on page 15

2.1

Jan 2022 P2H

Use the graph of $y = 2x^2 - 4x - 1$ to solve $2x^2 - 4x - 1 = 0$ (1 d.p.).

Answer:

2.2

Nov 2025 P2H

The graph of $y = \frac{1}{2}\left(x + \frac{4}{x}\right)$ is drawn. Which line solves $x + \frac{4}{x} = 6$? Solutions?

Answer:

2.3

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The graph of $y = x^2 - 3x$ is drawn. Which line would you draw to solve $x^2 - 4x + 1 = 0$?

Answer:

V3**VARIANT 3****Recognise the Shape**

Read the highest power of x and the sign — then match the shape.

SEEN IN PAST PAPERS

11 / 40

PRACTICE QUESTIONS: 3



Your notes

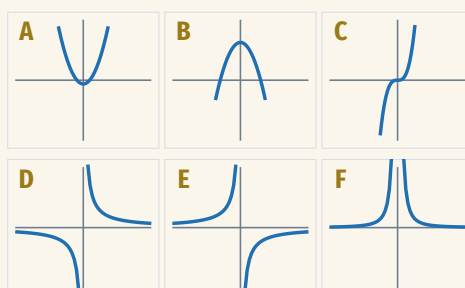
Came up in: Jan 2020 P2H Q15 • Jun 2022 P2HR Q16 • Jun 2023 P2HR Q12 • Jun 2023 P2H Q18 • Nov 2023 P1H Q12(c) • Jun 2024 P2HR Q16 • Jun 2025 P2H Q14

? What is it?**In plain words**

x^2 : **U** (or **n** if negative).

x^3 : **S-shape**. $\frac{1}{x}$: two pieces in opposite corners. $\frac{1}{x^2}$: two pieces, **both above**.

A minus sign **flips** the shape.

**How do I do it?****1 Power**

Highest power: 1 line, 2 parabola, 3 cubic.

2 Bottom?

x on the bottom: reciprocal (never touches the axes).

3 Sign

Negative in front: flip upside down.

4 Check

Put $x = 0$ (or $x = 1$) in: does the graph agree?

POWER, BOTTOM, SIGN Three questions name any shape.

V3 Recognise the Shape

VARIANT 3 • SEE IT DONE, THEN TRY

 Your notes
 WHEN YOU SEE → YOU DO

“Write down the letter of the graph that could have the equation”

→ Power, bottom, sign

$$y = -\frac{3}{x}$$

→ Reciprocal, flipped: top-left and bottom-right

$$y = \tan x^\circ, y = \sin x^\circ$$

→ Trig shapes (V5)

**WORKED EXAMPLE****PAST PAPER**

June 2023 • Paper 2HR • Q12 (shapes as in the gallery) • 3

marks

Match each equation to a shape in the gallery: (a) $y = \frac{10}{x^2}$ (b) $y = -\frac{3}{x}$
(c) $y = 5 - x^2$.

STEP 1 $\frac{10}{x^2} > 0$ always

◀ (a) two pieces, both above: F

STEP 2 $-\frac{3}{x} : x > 0 \Rightarrow y < 0$

◀ (b) reciprocal flipped: E

STEP 3 $-x^2 + 5$

◀ (c) negative x^2 : n, B**ANSWER** (a) F (b) E (c) B✓ $x = 1$: 10, -3, 4 ✓**WATCH OUT**

✗ $y = -\frac{3}{x}$ is D

✓ E

The minus flips it into the other two corners.

**EXAM TIP**

Test one value: $x = 1$ in $y = -\frac{3}{x}$ gives -3 — the graph must be below the axis there.

**Check yourself**

calculator first • answers on page 15

3.1

Jun 2025 P2H

Which gallery shape could be $y = 2x^3$?

Answer:

3.2

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Which gallery shape could be $y = \frac{4}{x}$?

Answer:

3.3

Dr Eslam

Which gallery shape could be $y = x^2 - 4$?

Answer:

V4

VARIANT 4

Sketch a Quadratic

Complete the square for the turning point;
 $x = 0$ for the y -intercept; $y = 0$ for the roots.

SEEN IN PAST PAPERS

2 / 40

PRACTICE QUESTIONS: 9

 Your notes

Came up in: Jan 2020 P1H Q20 • Nov 2024 P1H Q23

? What is it?

In plain words

A **sketch** is not a plot: show the shape and the **key points** only.

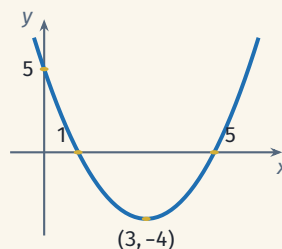
$y = (x - 3)^2 - 4$: turning point $(3, -4)$.

$x = 0$: **y-intercept**. $y = 0$: **roots**.

IT LOOKS LIKE

$$y = x^2 - 6x + 5$$

$$y = 4(x - 1)^2 - a$$



How do I do it?

1 Turning point

Complete the square: $(x - p)^2 + q \rightarrow (p, q)$.

2 y-axis

Put $x = 0$.

3 x-axis

Put $y = 0$; factorise or solve.

4 Sketch

U (or n) through the points; label them.

TP • x = 0 • y = 0 Label every point you found.

V4 Sketch a Quadratic

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Sketch the curve, showing the turning point" → **Complete the square**

"points of intersection with the coordinate axes" → **$x = 0$ and $y = 0$**

"The line $y = k$ meets the curve at (p, k) , (q, k) " → **p, q are roots of curve = k**

WORKED EXAMPLE DR ESAM

Sketch · 4 marks

Sketch $y = x^2 - 6x + 5$, showing the turning point and where it meets the axes.

STEP 1 $x^2 - 6x + 5 = (x - 3)^2 - 4$ ◀ complete the square

STEP 2 TP (3, -4) ◀ minimum (positive x^2)

STEP 3 $x = 0 : (0, 5)$ ◀ y-intercept

STEP 4 $(x - 1)(x - 5) = 0 : (1, 0), (5, 0)$ ◀ roots

ANSWER U-curve: TP (3, -4), (0, 5), (1, 0), (5, 0) ✓ TP halfway between the roots ✓

WATCH OUT

× TP (-3, -4)

✓ (3, -4)

$(x - 3)^2$ is smallest at $x = +3$.

EXAM TIP

The turning point sits exactly halfway between the roots — a free check.

Check yourself

calculator first · answers on page 15

4.1 Jan 2020 P1H

$y = 4(x - 1)^2 - a$, $a > 4$.
Give the turning point
and the y-intercept.

Answer:

4.2 Nov 2024 P1H

$y = x^2 - 8x - 9$ meets
 $y = k$ at (p, k) and (q, k) ,
 $p - q = 14$. Find k .

Answer:

4.3 Dr Eslam

Find the turning point
of $y = x^2 - 4x + 1$.

Answer:

V5

VARIANT 5

Trig Graphs

Know sin, cos, tan by heart; then read the stretch (a), the shift (b) and the lift (c).

SEEN IN PAST PAPERS

6 / 40

PRACTICE QUESTIONS: 4

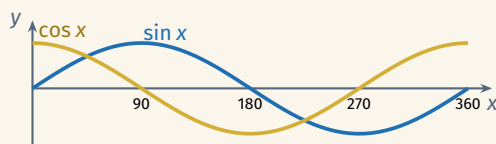
 Your notes

Came up in: Jun 2022 P2H Q23 • Jun 2023 P2HR Q26 • Jun 2024 P2H Q25 • Jun 2025 P2HR Q25

? What is it?**In plain words**

$\sin x$: starts at 0, top at 90° . $\cos x$: starts at 1, bottom at 180° . Both repeat every 360° .

$y = a \sin(x + b) + c$: a = height above the middle; c = middle line; b slides it left. $\cos bx$ repeats every $\frac{360}{b}$.

**IT LOOKS LIKE**

$$2 \sin(x + 60^\circ)$$

$$\cos\left(\frac{x}{2}\right)$$

$$a \cos bx + c$$

How do I do it?**1 Middle c**

$$\frac{\max + \min}{2}$$

2 Height a

$$\frac{\max - \min}{2}$$

3 Period / shift

Period $\frac{360}{b}$; or where the peak moved.

4 Points

Peak where the bracket = 90 (sin) or 0 (cos).

MIDDLE, HEIGHT, SHIFT Max and min give a and c .

V5 Trig Graphs

VARIANT 5 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"Find the coordinates of A" (a peak) → **Bracket = 90° for sin**

"Find a, b and c" → **c middle, a half the height, b from the period or shift**

$\cos\left(\frac{x}{2}\right)$ → **Stretched: repeats every 720°**

WORKED EXAMPLE **PAST PAPER**

June 2024 • Paper 2H • Q25 • 2 marks

The graph of $y = 2 \sin(x + 60^\circ)$ is sketched. A is the first maximum point; B is where the curve next crosses the x-axis going down. Find A and B.

STEP 1 $x + 60 = 90 \Rightarrow x = 30$ ◀ sin peaks at 90°

STEP 2 $A(30, 2)$ ◀ height $a = 2$

STEP 3 $x + 60 = 180 \Rightarrow x = 120$ ◀ sin is 0 again at 180°

STEP 4 $B(120, 0)$ ◀ on the axis

ANSWER $A(30, 2), B(120, 0)$ ✓ $2 \sin 180^\circ = 0$ ✓**WATCH OUT**✗ $x + 60 = 90$ so $x = 150$ ✓ $x = 30$

Subtract the 60, don't add it: the graph moved left.

EXAM TIP

Write the bracket = 90 or = 180 first — the coordinate follows.

Check yourself

calculator first • answers on page 15

5.1

Jun 2022 P2H

$y = \cos\left(\frac{x}{2}\right)$: A first crossing of the x-axis, B the minimum. Find A and B.

Answer:

5.2

Jun 2023 P2HR

$y = a \cos bx^\circ + c$: max 3, min -1 , peaks every 90° . Find a, b, c.

Answer:

5.3

Jun 2025 P2HR

$y = a \sin(x + b)^\circ + c$: max 4 at $x = 45$, min -2 . Find a, b, c.

Answer:

CHECK YOURSELF

Answers

Every answer below was checked with sympy (check_answers.py).

1.1 8, -2, -4, -2, 2

1.2 12, 3, 1.5, 1.2

1.3 -1, 3, 1, 3

2.1 $x \approx -0.2, 2.2$

2.2 $y = 3$; $x \approx 0.8, 5.2$

2.3 $y = x - 1$

3.1 C

3.2 D

3.3 A

4.1 TP (1, -a); (0, 4 - a)

4.2 $k = 24$

4.3 (2, -3)

5.1 A(180, 0), B(360, -1)

5.2 $a = 2, b = 4, c = 1$

5.3 $a = 3, b = 45, c = 1$

THE WHOLE TOPIC ON ONE PAGE

Graphs cheat sheet



Your notes
V1 Table, curve 18/40

$$(-2)^2 = 4$$

Bracket negatives; one smooth curve.

V2 Solve 3/40

$$2x^2 - 4x - 1 = 1 - 2x$$

Curve = line; read x.

V3 Shapes 11/40

$$-\frac{3}{x} \rightarrow E$$

Power, bottom, sign.

V4 Sketch 2/40

$$(x - 3)^2 - 4$$

TP, $x = 0$, $y = 0$.**V5 Trig 6/40**

$$a \sin(x + b) + c$$

Middle, height, shift.

**TOP 3 LOST MARKS**

1. $-2^2 = -4$ in a table ×
2. ruled or pointed curves ×
3. giving y for the solutions ×

BOTTOM LINE Bracket negatives • smooth curve • curve = line • power, bottom, sign.